Industrial Internship Report on

”Forecasting of Smart City Traffic Patterns”

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Executive Summary
This report describes the industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Unicoverage Technologies Pvt Ltd. The Internship focused on solving a real-world problem related to smart city development. My project was “Forecasting of smart City Traffic Patterns”. The objective of this project was to analyze traffic data from different junctions in a smart city and predict traffic congestion patterns using data analysis and machine learning techniques. During internship, I worked on understanding traffic datasets, preprocessing the data, and applying forecasting techniques to predict future traffic conditions. The project helps government authorities plan better traffic management systems and improve infrastructure of machine learning models.

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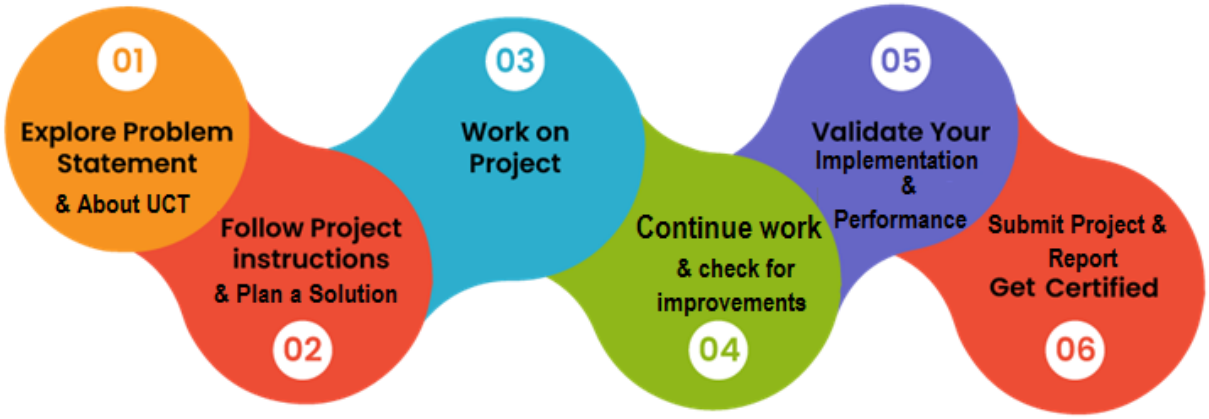
1 Preface

The internship program was conducted for six weeks and aimed to provide practical exposure to real-world industry problems.

The project assigned to me was Forecasting of Smart City Traffic Patterns, which focuses on predicting traffic flow in different city junctions using historical traffic data.

Through this internship, I gained hands-on experience in data analysis, data preprocessing, and predictive modeling. I also learned how machine learning techniques can be applied to solve real-world problems.

How Program was planned:



I would like to thank Upskill Campus, Unicoverage Technologies Pvt Ltd, and The IoT Academy for providing this opportunity and guidance throughout the internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

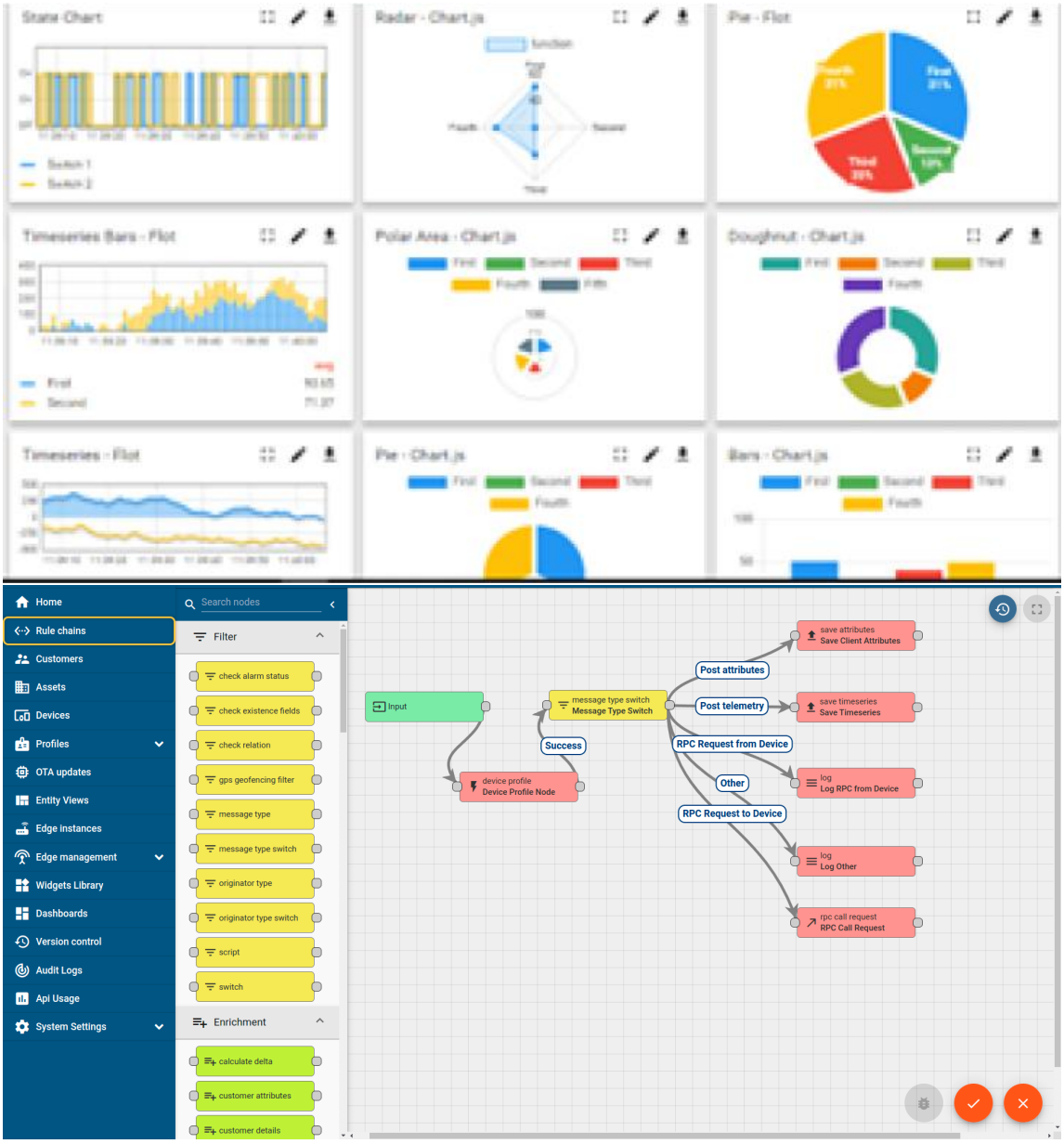
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

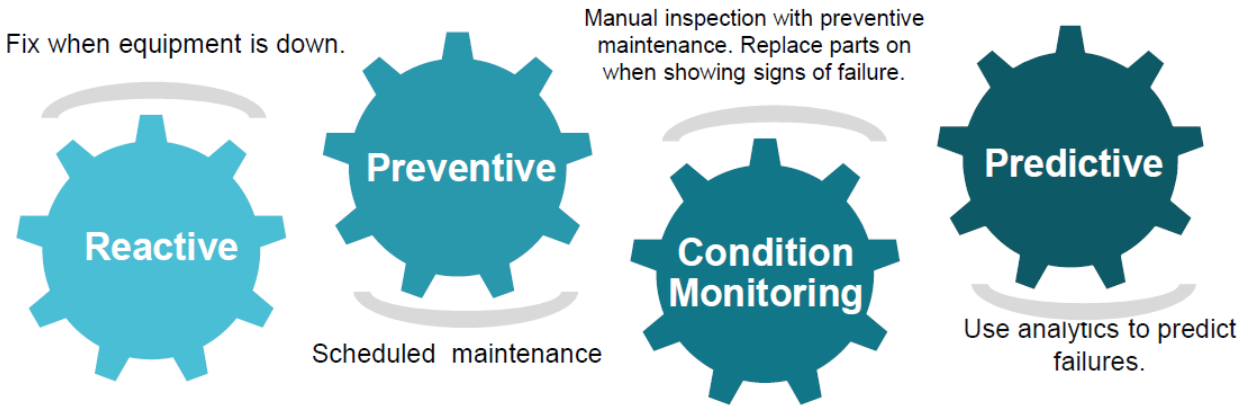
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

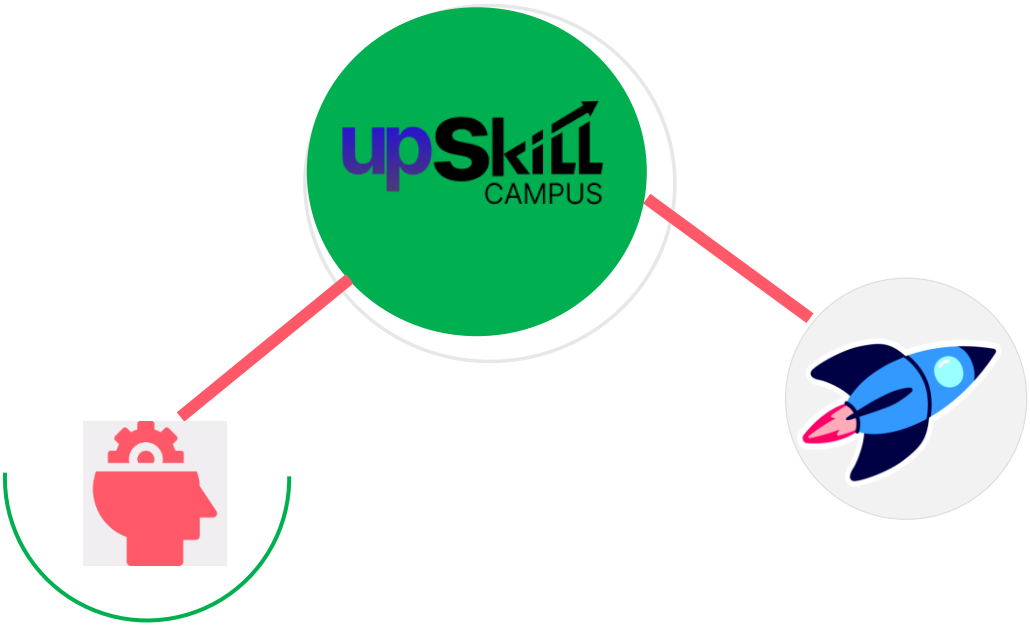


2.2 About upskill Campus (USC)

Upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way. It is a career development platform that provides students with opportunities to enhance their technical and professional skills. It offers internships,

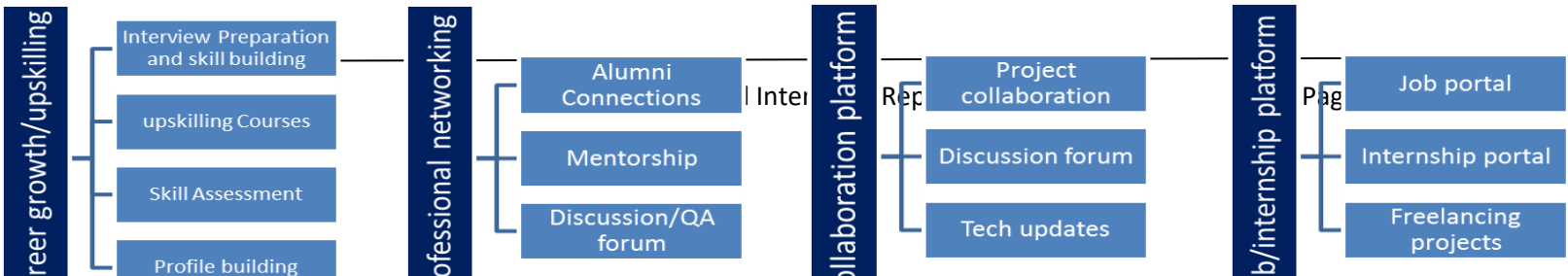
Industry projects, Mentorship, and career guidance to help students become industry-ready.



Seeing need of up skilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

Upskill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Smart City Traffic Dataset

[2] Machine Learning and Data Analysis resources

[3] Online Documentation and research papers related to traffic prediction

2.6 Glossary

Terms	Acronym
Smart city	A city that uses technology and data to improve urban services
Traffic Forecasting	Predicting traffic conditions using historical data
Machine Learning	Algorithms that learn patterns from data
Data Analysis	Process of examining data to extract useful insights

3 Problem Statement

Smart Cities aim to improve urban infrastructure and services using digital technologies. One of the major challenges faced by cities is traffic congestion.

The government wants to develop a system that can predict traffic patterns across four major city junctions. Traffic patterns vary during working days, holidays, and special events.

The Objective of this project is to analyze historical traffic data and build a model that can forecast future traffic flow. This will help authorities in better traffic planning, congestion management, and infrastructure development.

4 Existing and Proposed solution

EXISTING SOLUTIONS

Many traffic management systems rely on fixed traffic signals or manual monitoring. These systems often fail to adapt to changing traffic conditions.

Limitation includes:

-Lack of real-time analysis

-Inefficient traffic signal timing

-Limited forecasting capabilities

PROPOSED SOLUTION

The proposed solution uses data analysis and machine learning models to forecast traffic patterns.

The System will:

-Analyzes historical traffic data

-Identify traffic trends and patterns

-Predict future traffic congestion levels

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram :

Traffic Data -> Data Preprocessing -> Feature Engineering -> Machine learning Model -> Traffic Forecast output

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram:

Data collection -> Data Cleaning -> Data Transformation -> Model Training -> Model Evaluation ->Prediction Generation

Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces :

- Dataset input (CSV traffic file)
- Data Processing using Python
- Model implementation using machine learning libraries

6 Performance Test

6.1 Test Plan/ Test Cases

The model was tested using historical traffic datasets. The dataset was divided into training and testing data to evaluate prediction accuracy.

6.2 Test Procedure

1. Import dataset
2. Preprocess the data
3. Train the forecasting model
4. Test predictions on unseen data
5. Evaluate accuracy

6.3 Performance Outcome

The model successfully identified traffic patterns and produced predictions for future traffic flow. The results can help authorities anticipate congestion and take preventive actions.

7 My learnings

During this internship, I learned many important concepts including

- ✓ Data Preprocessing
- ✓ Traffic data analysis
- ✓ Machine learning model implementation
- ✓ Problem solving and analytical thinking
- ✓ Importance of data in smart city development

This internship helped me Understand how technology and data science can solve real-world urban problems.

7 Future work scope

Future improvements for this project mat include:

- Using real-World traffic Sensor data
- Implementing advanced deep learning models
- Integrating the system with smart traffic signals
- Developing a dashboard for visualizing traffic predictions.

These improvements can make the system more accurate and useful for smart city management .